Module 07 – Maximal Flow

Model Formulation

1. Maximize maximal flow by changing the units of flow
2. Subject to the constraints of:

Units of flow >=0
Units of flow <= Upper bounds
Net flow – Supply/Demand

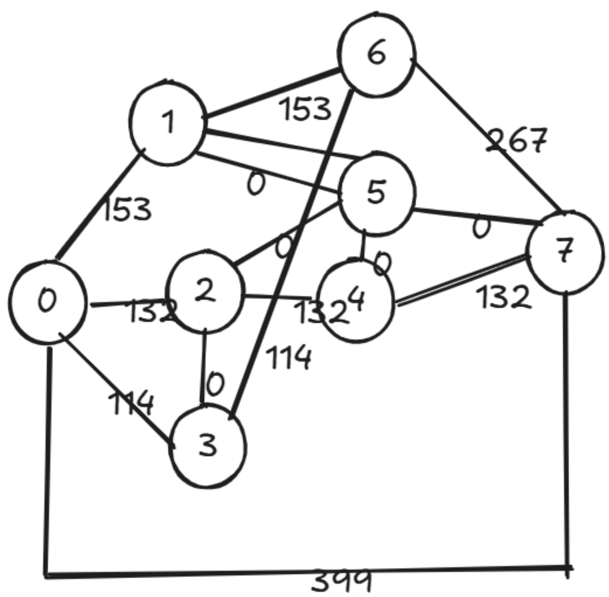
Model Optimized for Maximal Flow

Maximal Flow -> 399

Units of Flow	Links From		To		Upper Bound
153	0	Butter Pecan Bluff	1	Fruity Gusher Geyser	204
132	0	Butter Pecan Bluff	2	Gingerbread Glades	132
114	0	Butter Pecan Bluff	3	Jelly River Delta	121
153	1	Fruity Gusher Geyser	6	Strawberry Swirl Stream	237
0	2	Gingerbread Glades	3	Jelly River Delta	270
132	2	Gingerbread Glades	4	Pixie Stix Plateau	179
0	2	Gingerbread Glades	5	Pudding Peaks	223
114	3	Jelly River Delta	6	Strawberry Swirl Stream	114
132	4	Pixie Stix Plateau	7	Toffee Town	243
0	4	Pixie Stix Plateau	5	Pudding Peaks	194
0	5	Pudding Peaks	7	Toffee Town	108
0	5	Pudding Peaks	1	Fruity Gusher Geyser	285
267	6	Strawberry Swirl Stream	7	Toffee Town	267
399	7	Toffee Town	0	Butter Pecan Bluff	9999

Nodes		Inflow	Outflow	Net Flow	Supply / Demand
0	Butter Pecan Bluff	399	399	0	**0**
1	Fruity Gusher Geyser	153	153	0	**0**
2	Gingerbread Glades	132	132	0	**0**
3	Jelly River Delta	114	114	0	**0**
4	Pixie Stix Plateau	132	132	0	**0**
5	Pudding Peaks	0	0	0	**0**
6	Strawberry Swirl Stream	267	267	0	**0**
7	Toffee Town	399	399	0	**0**

This is a maximum flow problem where the goal is to send as much flow as possible from the source (Butter Pecan Bluff, node 0) to the sink (Toffee Town, node 7) while respecting the capacity constraints on each link and maintaining flow balance at intermediate nodes. In this case, the bottleneck edges are:
From (1,6), (2,3), (4,5), (5,7) and (5,1)



Model with Stipulation

Node	Inflow (units reaching)
0	399
1	153
2	132
3	114
4	132
5	0
6	267
7	399